Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CE743]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

TRAFFIC ENGINEERING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Explain in detail both the traffic characteristics.	**10**
	b.	Explain PIEV theory.	**04**
2.	**a.**	Explain in brief Level of service concept and factors affecting LOS.	**07**
	b.	Define the term spot speed. Explain the presentation of spot speed data.	**07**
3.	**a.**	Explain 2 phase signal and 4 phase signal.	**08**
	b.	Explain three types of traffic signs with examples for each with neat diagram.	**06**
		OR	
4.	**a.**	What is the need of street lighting? Explain any two terminologies used in street lighting.	**07**
	b.	List the design factors of street lighting. Explain any three types.	**07**
5.	**a.**	Describe the causes of road accidents and also suggest preventive measures to control accidents.	**08**
	b.	Explain the importance and promotion of public and non motorized transport.	**06**
		OR	
6.	**a.**	Explain the measures to control the traffic noise.	**08**
	b.	How to control air pollution due to traffic?	**06**
7.	**a.**	List the traffic management measures. Explain any three in detail	**10**
	b.	List out various travel demand management techniques.	**04**
		OR	
8.	**a.**	Define ITS. What are the applications of ITS in traffic engineering?	**05**
	b.	Write in detail any three methods of forecasting in traffic volume prediction.	**09**